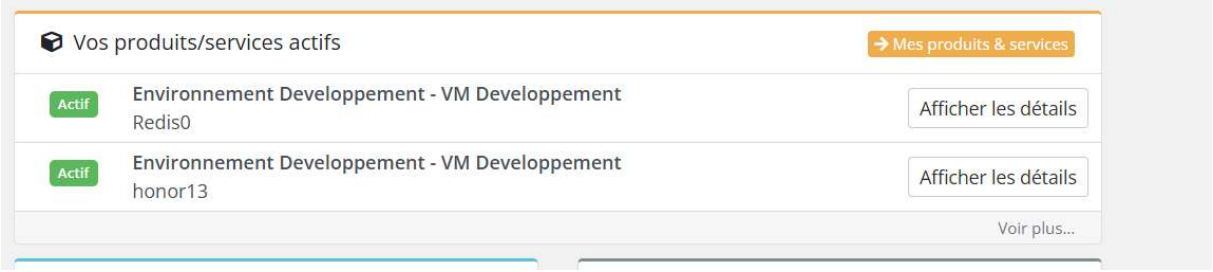
Documentation REDIS Cluster

Introduction

Redis est une base de données NoSQL clé-valeur largement utilisée pour des cas d'usage tels que la mise en cache, la gestion de sessions et les files de messages. Dans le cadre d'une architecture micro-services ou web hautement performants, Redis doit être **hautement disponible**, **résilient aux pannes**, et capable de **monter en charge horizontalement**. Ce document décrit le déploiement d'une solution Redis résiliente et scalable en utilisant des outils comme **HAProxy**, **Docker**, et une architecture de type **Redis Cluster** pour garantir la haute disponibilité et la répartition de la charge.

Mise en place des prérequis :

1) VM MASTER ET BACKUP



Redis0 (ip : 10.0.50.103) sera ma VM Backup et honor13 (ip :10.0.50.149) ma VM Master

2) AJOUT D'UNE ADRESSE VIP SUR MA MACHINE MASTER

Nmap -sn 10.0.50.0/24 afin de scanner le réseau pour voir les ip déjà utilisés sur cette plage

```
Processing triggers for libc-bin (2.39-0ubuntu8.4) ...
root@honor13:~# nmap -sn 10.0.50.0/24
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-05-15 14:17 UTC
Nmap scan report for 10.0.50.1
Host is up (0.00049s latency).
MAC Address: 00:00:5E:00:01:DF (Icann, Iana Department)
Nmap scan report for 10.0.50.2
```

Toutes les adresses utilisées ont été lister je vais ainsi pouvoir en choisir une qui n'est pas occupée

```
Host is up.
Nmap done: 256 IP addresses (34 hosts up) scanned in 2.09 seconds
root@honor13:~#
```

Cette commande va permettre d'y ajouter l'adresse VIP de mon choix(10.0.50.4/24)

```
Nmap done: 256 IP addresses (34 hosts up) scanned in 2.09
root@honor13:~# sudo ip addr add 10.0.50.4/24 dev eth0
```

Mon adresse a bien été ajouté mais il faudra la faire en sorte que personne d'autre ne l'utilise avec **arping -I eth0 (ton adresse ip)**

```
root@honor13:~# ip a
ping 10.0.50.200  # depuis une autre VM
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue s
en 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0@if558: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 15
up default qlen 1000
    link/ether bc:24:11:5f:e4:0e brd ff:ff:ff:ff:ff:ff
    inet 10.0.50.149/24 brd 10.0.50.255 scope global et
       valid_lft forever preferred_lft forever
    inet 10.0.50.4/24 scope global secondary eth0
       valid_lft forever preferred_lft forever
```

3) INSTALLATION DE DOCKER ET DE DOCKER COMPOSE SUR TON SYSTEME

Pas besoin d'installer Docker il sont déjà prédéfini sur WHMCS

```
root@honor13:~#
root@honor13:~# docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS   PORTS   NAMES
root@honor13:~#
```

4°) VERIFICATION DES PORTS 6379, 6384, 80 ET 443

Tous les ports sont libres en dehors du port 80. on peut le remplacer par 81

```
Ports
----------------------------

0.0.0.0:6382->6379/tcp,:::6382->6379/tcp
0.0.0.0:6383->6379/tcp,:::6383->6379/tcp
0.0.0.0:6384->6379/tcp,:::6384->6379/tcp
0.0.0.0:6379->6379/tcp,:::6379->6379/tcp
0.0.0.0:6380->6379/tcp,:::6380->6379/tcp
0.0.0.0:6381->6379/tcp,:::6381->6379/tcp
```

Étapes d'installation Redis Cluster

Pour l'étape 1, on va créer le nœuds de redis cluster et tester sa fonctionnalité:

A°)Répertoire les fichiers sur VM Master :

Redis.conf

```
  GNU nano 7.2
# Network
port 6379
bind 0.0.0.0
protected-mode no

# Cluster
cluster-enabled yes
cluster-config-file nodes.conf
cluster-node-timeout 5000
cluster-require-full-coverage no
cluster-allow-reads-when-down yes

# Replication
repl-diskless-sync yes
repl-backlog-size 1gb
repl-timeout 60
min-replicas-to-write 1
min-replicas-max-lag 20
cluster-replica-validity-factor 0

# Persistence
appendonly yes
appendfsync everysec
aof-rewrite-incremental-fsync yes
daemonize no

# Memory
maxmemory 1gb
maxmemory-policy allkeys-lru

# Performance
tcp-keepalive 60
```

B°) docker-compose.yml sur la VM Master et Backup

```
  GNU nano 7.2                    docker-compose.yml
version: "3.9"

services:
  redis-1:
    image: redis:latest
    container_name: redis1
    command: redis-server /usr/local/etc/redis/redis.conf
    volumes:
      - redis-data-1:/data
      - ./redis.conf:/usr/local/etc/redis/redis.conf
    ports:
      - "6379:6379"
    networks:
      - redis-cluster-net

  redis-2:
    image: redis:latest
    container_name: redis2
    command: redis-server /usr/local/etc/redis/redis.conf
    volumes:
      - redis-data-2:/data
      - ./redis.conf:/usr/local/etc/redis/redis.conf
    ports:
      - "6380:6379"
    networks:
      - redis-cluster-net

  redis-3:
    image: redis:latest
    container_name: redis3
    command: redis-server /usr/local/etc/redis/redis.conf
    volumes:
      - redis-data-3:/data
      - ./redis.conf:/usr/local/etc/redis/redis.conf
    ports:
      - "6381:6379"
    networks:
      - redis-cluster-net

  redis-replica-1:
    image: redis:latest
    container_name: redis-replica-1
    command: redis-server /usr/local/etc/redis/redis.conf

^G Help      ^O Write Out  ^W Where Is   ^K Cut       ^T Execute   ^C Location
```

Avec les 3 clusters et leurs réplicas

Démarrage du cluster avec docker-compose up -d et vérification du cluster

```
root@honor13:/home/redis/redis-server# docker-compose ps
       Name                     Command            State              Ports
-------------------------------------------------------------------------------
redis-cluster-setup   sh -c sleep 10 && redis-    Exit 0
                      cl ...
redis-replica-1       docker-entrypoint.sh        Up       0.0.0.0:6382-
                      redis ...                            >6379/tcp,:::6382-
                                                           >6379/tcp
redis-replica-2       docker-entrypoint.sh        Up       0.0.0.0:6383-
                      redis ...                            >6379/tcp,:::6383-
                                                           >6379/tcp
redis-replica-3       docker-entrypoint.sh        Up       0.0.0.0:6384-
                      redis ...                            >6379/tcp,:::6384-
                                                           >6379/tcp
redis1                docker-entrypoint.sh        Up       0.0.0.0:6379-
                      redis ...                            >6379/tcp,:::6379-
                                                           >6379/tcp
redis2                docker-entrypoint.sh        Up       0.0.0.0:6380-
                      redis ...                            >6379/tcp,:::6380-
                                                           >6379/tcp
redis3                docker-entrypoint.sh        Up       0.0.0.0:6381-
                      redis ...                            >6379/tcp,:::6381-
                                                           >6379/tcp
```

Plan de test unitaire

```
root@honor13:/home/redis/redis-server# docker exec -it redis1 redis-cli cluster node
s
eafd0eecd91c2554c5fdd716c4f4e93dd3bb1bd8 172.18.0.2:6379@16379 myself,master - 0 0 1
 connected 0-5460
dd1f849cc3ee2faf39cb6d4a751ac4d7a4dc66be 172.18.0.3:6379@16379 master - 0 1747377540
838 2 connected 5461-10922
822c775e4700e341e8a4aef0627852890fc2c5cc 172.18.0.4:6379@16379 master - 0 1747377541
844 3 connected 10923-16383
74389f8bf217adad08617ca2d278f5bc5e24818e 172.18.0.7:6379@16379 slave dd1f849cc3ee2fa
f39cb6d4a751ac4d7a4dc66be 0 1747377540533 2 connected
181aac9251974c32a73877457b6bfd2484704a26 :0@0 slave,noaddr 822c775e4700e341e8a4aef06
27852890fc2c5cc 1747375900027 1747375899975 3 disconnected
a06bdf142d573f2c82cd24f2aed630d7b138b540 172.18.0.6:6379@16379 slave eafd0eecd91c255
4c5fdd716c4f4e93dd3bb1bd8 0 1747377541000 1 connected
root@honor13:/home/redis/redis-server# docker ps -a | grep 181aac9251974c32a7
```

Plan de test unitaire

1-Vérification du lancement des conteneurs ✅

```
root@honor13:/home/redis/redis-server# docker-compose ps
       Name                     Command            State              Ports
-------------------------------------------------------------------------------
redis-cluster-setup   sh -c sleep 10 && redis-    Exit 0
                      cl ...
redis-replica-1       docker-entrypoint.sh        Up       0.0.0.0:6382-
                      redis ...                            >6379/tcp,:::6382-
                                                           >6379/tcp
redis-replica-2       docker-entrypoint.sh        Up       0.0.0.0:6383-
                      redis ...                            >6379/tcp,:::6383-
                                                           >6379/tcp
redis-replica-3       docker-entrypoint.sh        Up       0.0.0.0:6384-
                      redis ...                            >6379/tcp,:::6384-
                                                           >6379/tcp
redis1                docker-entrypoint.sh        Up       0.0.0.0:6379-
                      redis ...                            >6379/tcp,:::6379-
                                                           >6379/tcp
redis2                docker-entrypoint.sh        Up       0.0.0.0:6380-
                      redis ...                            >6379/tcp,:::6380-
                                                           >6379/tcp
redis3                docker-entrypoint.sh        Up       0.0.0.0:6381-
                      redis ...                            >6379/tcp,:::6381-
                                                           >6379/tcp
```

2-Vérification de la formation du cluster Redis (sur VM1) ✅

```
root@honor13:/home/redis/redis-server# docker exec -it redis1 redis-cli cluster node
s
eafd0eecd91c2554c5fdd716c4f4e93dd3bb1bd8 172.18.0.2:6379@16379 myself,master - 0 0 1
 connected 0-5460
dd1f849cc3ee2faf39cb6d4a751ac4d7a4dc66be 172.18.0.3:6379@16379 master - 0 1747377540
838 2 connected 5461-10922
822c775e4700e341e8a4aef0627852890fc2c5cc 172.18.0.4:6379@16379 master - 0 1747377541
844 3 connected 10923-16383
74389f8bf217adad08617ca2d278f5bc5e24818e 172.18.0.7:6379@16379 slave dd1f849cc3ee2fa
f39cb6d4a751ac4d7a4dc66be 0 1747377540533 2 connected
181aac9251974c32a73877457b6bfd2484704a26 :0@0 slave,noaddr 822c775e4700e341e8a4aef06
27852890fc2c5cc 1747375900027 1747375899975 3 disconnected
a06bdf142d573f2c82cd24f2aed630d7b138b540 172.18.0.6:6379@16379 slave eafd0eecd91c255
4c5fdd716c4f4e93dd3bb1bd8 0 1747377541000 1 connected
root@honor13:/home/redis/redis-server# docker ps -a | grep 181aac9251974c32a7
```

3-Validation de la création du cluster Redis ✅

```
root@honor13:/home/redis/redis-server# docker exec -it redis1 redis-cli cluster slot
s
1) 1) (integer) 0
   2) (integer) 5460
   3) 1) "172.18.0.2"
      2) (integer) 6379
      3) "eafd0eecd91c2554c5fdd716c4f4e93dd3bb1bd8"
      4) (empty array)
   4) 1) "172.18.0.6"
      2) (integer) 6379
      3) "a06bdf142d573f2c82cd24f2aed630d7b138b540"
      4) (empty array)
2) 1) (integer) 5461
   2) (integer) 10922
   3) 1) "172.18.0.3"
      2) (integer) 6379
      3) "dd1f849cc3ee2faf39cb6d4a751ac4d7a4dc66be"
      4) (empty array)
   4) 1) "172.18.0.7"
      2) (integer) 6379
      3) "74389f8bf217adad08617ca2d278f5bc5e24818e"
      4) (empty array)
3) 1) (integer) 10923
   2) (integer) 16383
   3) 1) "172.18.0.4"
      2) (integer) 6379
      3) "822c775e4700e341e8a4aef0627852890fc2c5cc"
      4) (empty array)
```

Informations sur l'état du cluster ✅

```
root@honor13:/home/redis/redis-server# docker exec -it redis1 redis-cli cluster info
cluster_state:ok
cluster_slots_assigned:16384
cluster_slots_ok:16384
cluster_slots_pfail:0
cluster_slots_fail:0
cluster_known_nodes:6
cluster_size:3
cluster_current_epoch:6
cluster_my_epoch:1
cluster_stats_messages_ping_sent:6167
cluster_stats_messages_pong_sent:5796
cluster_stats_messages_fail_sent:12
cluster_stats_messages_sent:11975
cluster_stats_messages_ping_received:5796
cluster_stats_messages_pong_received:6162
cluster_stats_messages_fail_received:3
cluster_stats_messages_received:11961
total_cluster_links_buffer_limit_exceeded:0
```

Test de la réplication ✅

```
root@honor13:/home/redis/redis-server# docker exec -it redis2 redis-cli
tion
# Replication
role:master
connected_slaves:1
min_slaves_good_slaves:1
slave0:ip=172.18.0.7,port=6379,state=online,offset=5138,lag=0
master_failover_state:no-failover
master_replid:4d5532337a0269b3d643c7e2d2ea358980f00ee2
master_replid2:0000000000000000000000000000000000000000
master_repl_offset:5138
second_repl_offset:-1
repl_backlog_active:1
```

Test de connectivité et de routage automatique ✅

```
root@honor13:/home/redis/redis-server# docker exec redis1 redis-cli PING
PONG
root@honor13:/home/redis/redis-server# docker exec -it redis1 redis-cli -c set user1
 "Toto"
docker exec -it redis1 redis-cli -c get user1
OK
"Toto"
root@honor13:/home/redis/redis-server#
```

Simulation de panne ✅